

UNSUPERVISED MACHINE LEARNING · PROJECT REPORT

Unsupervised Phenotyping of Cardiac Risk

A Clustering & Bayesian Network approach to the UCI Heart Disease dataset

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Why look for cardiac phenotypes without labels?

Cardiovascular disease is the leading cause of mortality worldwide. Supervised classifiers need a diagnostic label — but in practice labels are often noisy, incomplete, or absent. This motivates an unsupervised question:

“Given only routine clinical measurements, can we discover natural patient groupings that correspond to interpretable cardiac phenotypes?”

Two objectives

- Partition the population with three structurally distinct clustering algorithms, rigorously validated.
- Learn a Bayesian Network over the clinical variables + discovered cluster label to expose conditional dependencies.

THE DATASET

UCI Heart Disease

Detrano et al., 1989 — a long-standing clinical benchmark spanning four collection sites.

920

patient records

15

raw attributes

4

clinical sites

0

labels used

Reading the data before touching it

EDA focused on schema, attribute cardinality and — critically — per-column missingness, which drove the whole cleaning strategy. Five continuous features (age, resting BP, cholesterol, max heart rate, ST-depression) are visibly skewed and span heterogeneous ranges.



Heterogeneous supports & variance

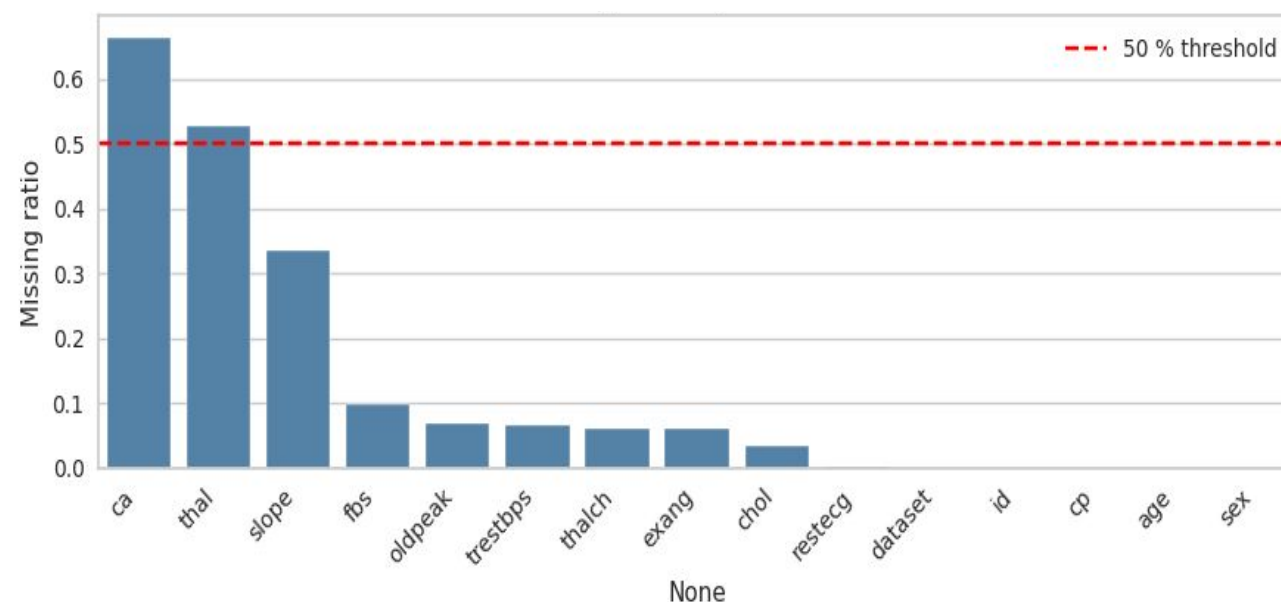
Differing ranges → median imputation + standardisation are the right choices.



Skewed continuous distributions

Robust, median-based statistics preferred over mean-based ones.

Missing-value ratio per attribute (%)



ca & thal exceed the 50% threshold → dropped as non-recoverable

A pipeline built for distance parity

1

Clean & drop

Remove id & dataset (no clinical signal); drop ca, thal (>50% missing).

2

Impute

Median for numeric, most-frequent (mode) for categorical — resistant to skew.

3

One-hot encode

drop_first=True avoids collinearity and false ordinal distances.

4

Standardise

StandardScaler → zero mean, unit variance so no feature dominates.

5

Isolation Forest

Contamination 0.05 flags 46 anomalies (~5%), removed before clustering.



The scaling problem

In the raw data, the variance of serum cholesterol is roughly two orders of magnitude larger than that of age.

Without standardisation, Euclidean distance is dominated by the highest-variance axis and the partition degenerates into a univariate split on cholesterol.

Quantified experimentally in Section VI →

Escaping the curse of dimensionality

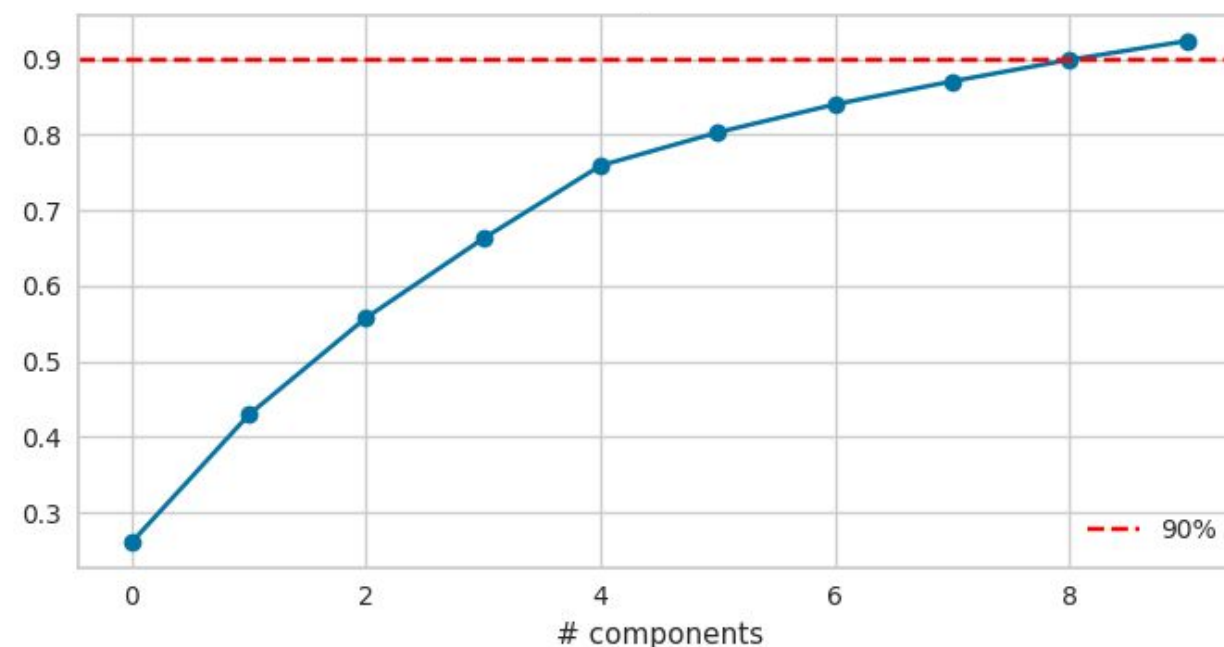
One-hot encoding inflates the feature space. In high dimensions, Euclidean distances concentrate — the contrast between a point's nearest and farthest neighbour vanishes, undermining any distance-based notion of a cluster.

19 → **10**

PCA retains 90% of total variance, compressing the encoded 19-D space to 10 principal components. Clustering runs in this reduced space; noise directions are discarded.

t-SNE (perplexity 20) is used only as a 2-D visual aid — never as the input geometry for clustering.

Cumulative explained variance



10 components cross the 90% threshold

Three inductive biases, one verdict

Three structurally distinct families were chosen deliberately — agreement between them is stronger evidence of genuine structure than any single method could give.



CENTROID

K-means++

Assumes convex, isotropic clusters; needs k in advance. Elbow + silhouette chosen jointly.

Seeded with k -means++, 10 restarts (lowest SSE).



HIERARCHICAL

Agglomerative (Ward)

Variance-minimising merges; no distributional assumption. Reveals nested structure.

Cut chosen at the largest dendrogram gap.



DENSITY

DBSCAN

No prior k ; recovers arbitrary shapes and labels sparse points as noise.

ϵ from the k -distance knee, refined by grid search.

Optimal clustering by internal validation

Silhouette (higher is better) is the primary metric, cross-checked with the **Davies–Bouldin** index (lower is better). No ground-truth labels are ever used.



DBSCAN wins

It produces the cleanest, most stable partition — isolating the dense core and relegating diffuse points to noise.

0.40

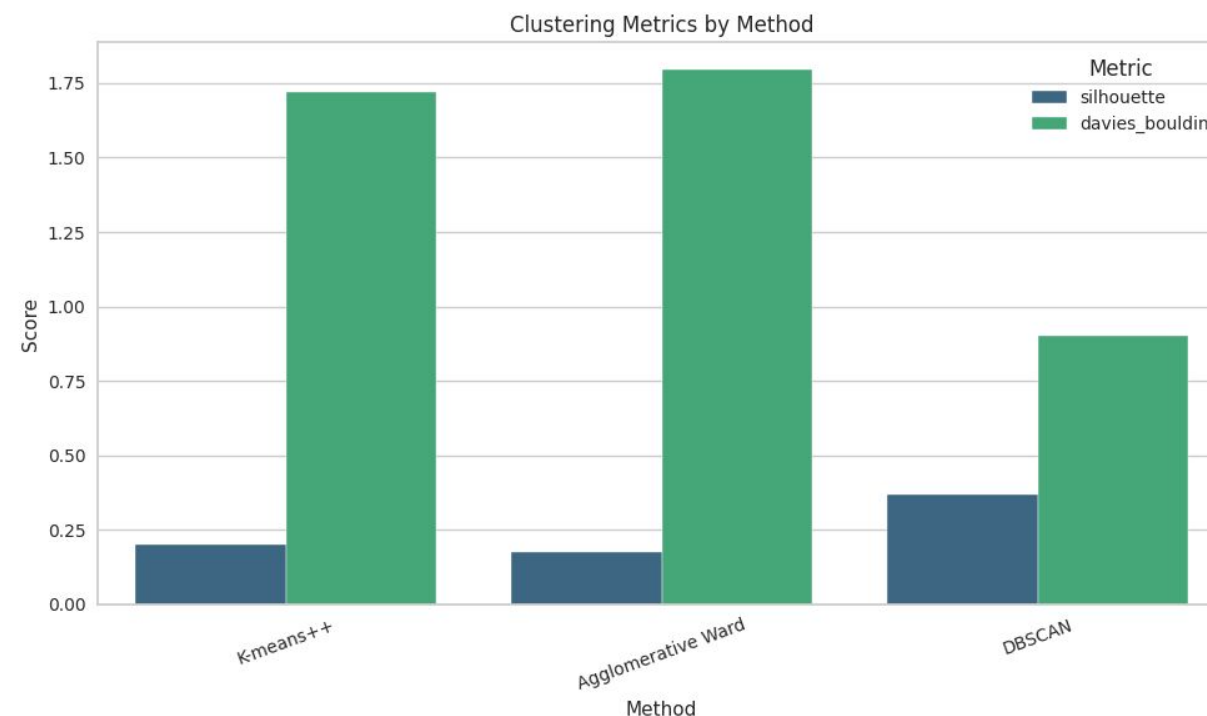
silhouette

0.90

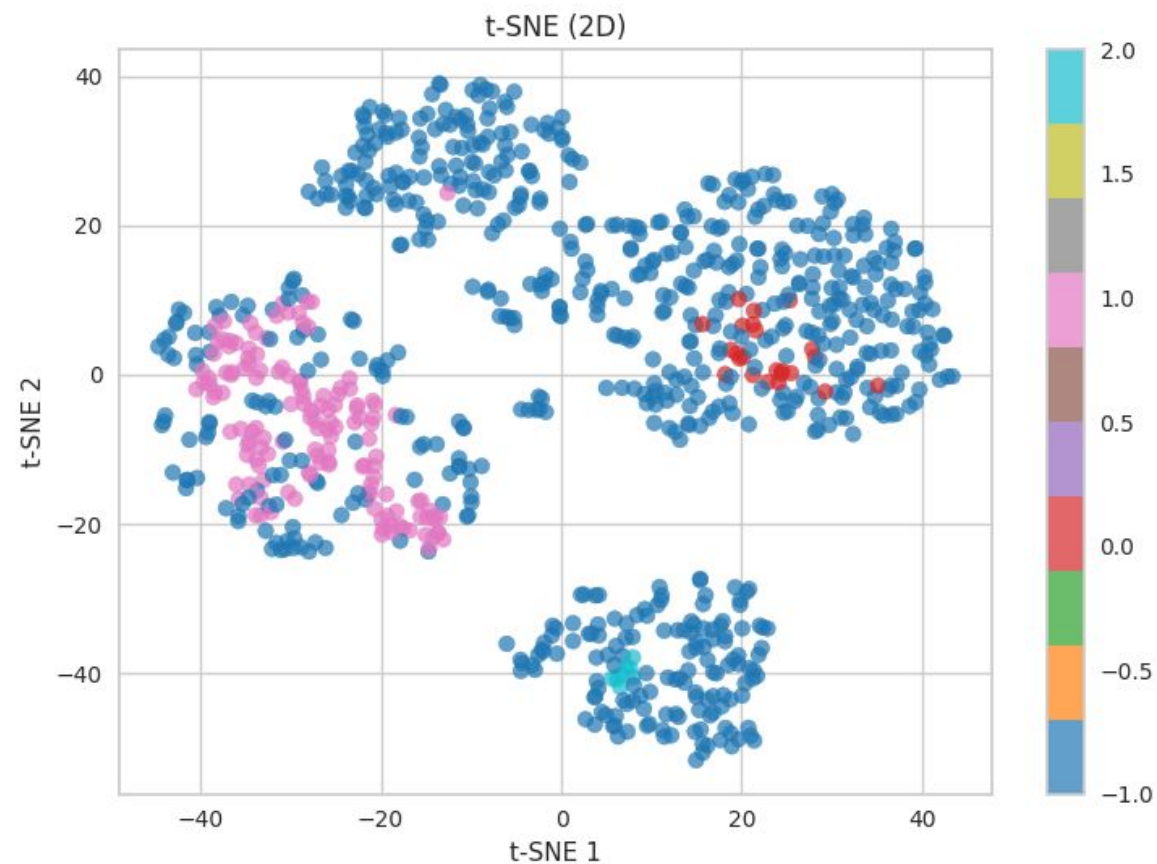
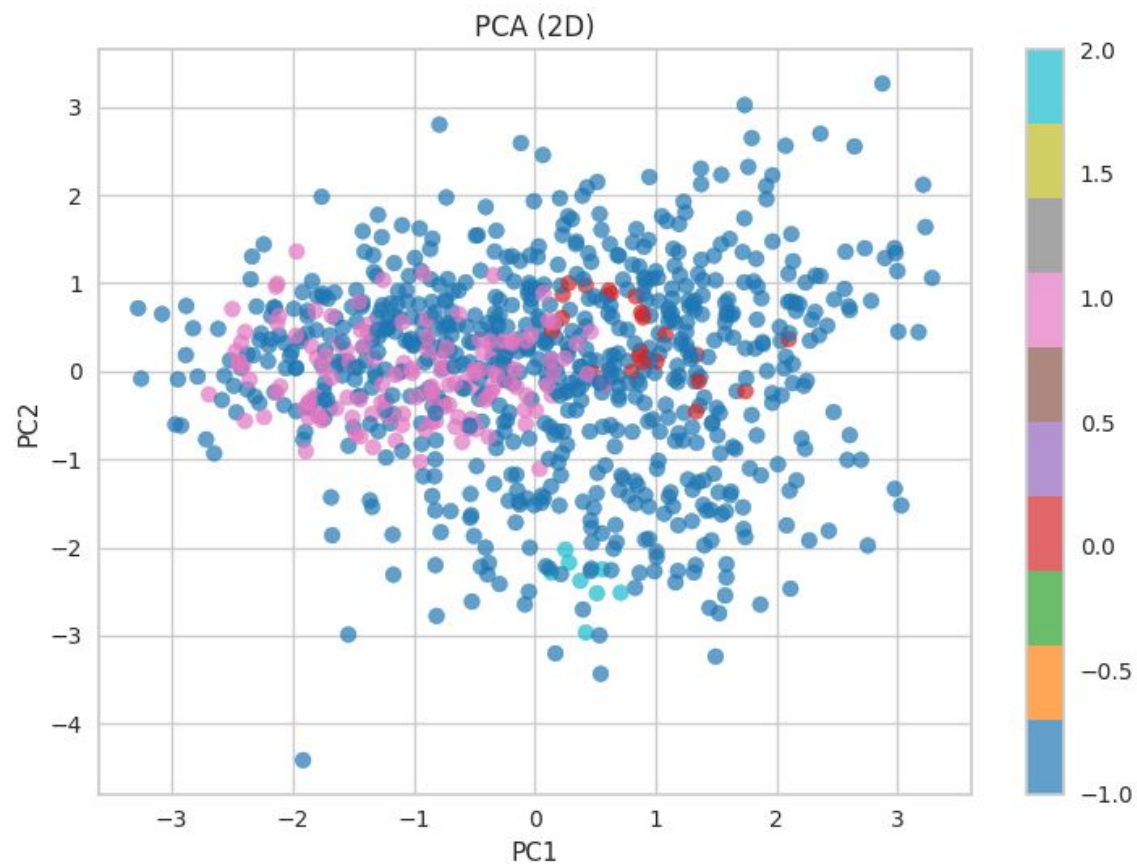
Davies–Bouldin

K-means++ & Ward agree on a common $k = 3$ solution (silhouette ≈ 0.15 – 0.16) — mixed-type clinical data rarely forms widely separated spherical clouds.

Internal validation on the standardised, PCA-reduced space



DBSCAN CLUSTERS 2D-VISUALIZATION



What if we don't scale the features?

K-means was run on three variants of the data. The unscaled matrix yields a deceptively high silhouette — but it is an artefact, not structure.

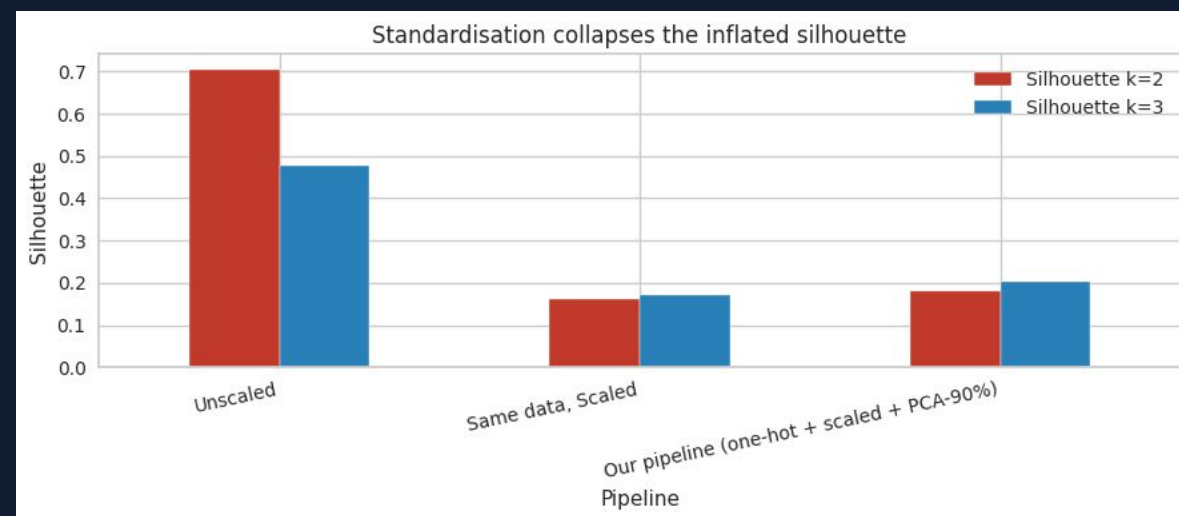


The cholesterol trap

Clustering on the raw chol column alone reproduces most of the inflated score — confirming the apparent quality was a univariate split on one dominant, unnormalised variable.

Standardised silhouettes (≈ 0.16) are the honest measure of clustering difficulty on this dataset.

K-means silhouette across pipelines



The cluster label as a latent risk stratifier

Continuous features are binned into 5 quantiles; the best K-means partition is appended as a Cluster node. A DAG is learned by Hill-Climb search, scored with BIC (max in-degree 3).

$$\text{BIC} = -2\ell(\theta) + k \cdot \log n$$

The $k \cdot \log n$ penalty is stricter than AIC's $2k$ (for $n > 7$) → a sparser, more interpretable DAG, ideal in a clinical setting.

Two mirror-image phenotypes

Silent ischaemia

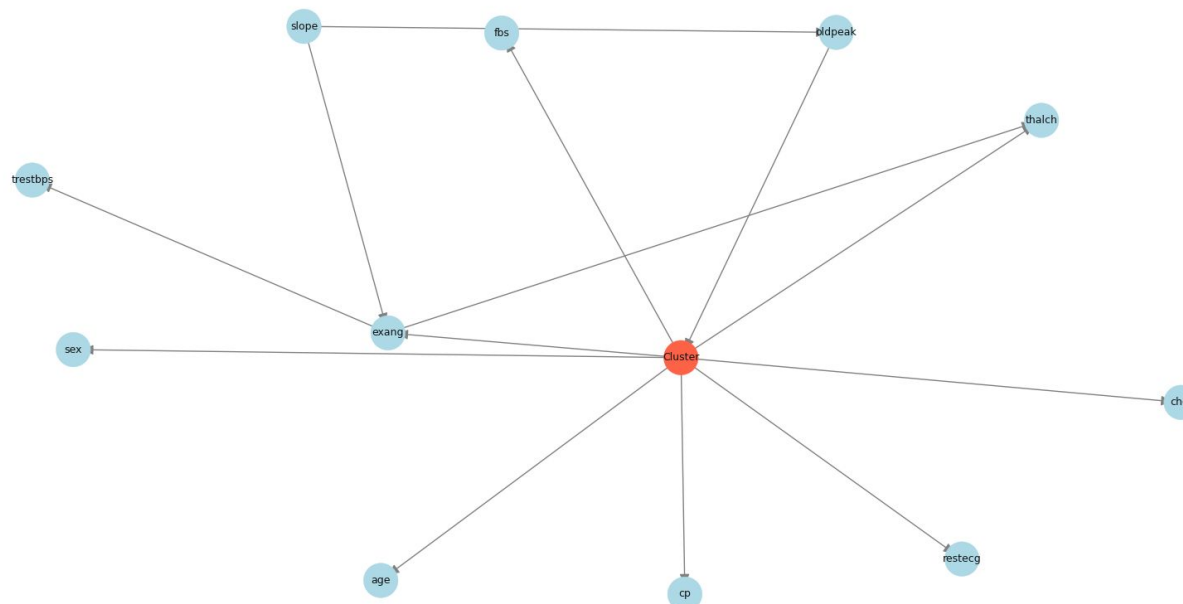
Asymptomatic chest pain, exercise-induced angina, reduced max heart rate — high burden.

Lower-risk

Typical/atypical angina, low angina probability, preserved heart rate.

Cluster is the most central node (betweenness)

Bayesian Network structure (BIC-scored Hill-Climb)



Structure learning gives the skeleton — the next step is querying it.

Querying the network: exact inference

With the DAG fixed, conditional probability tables are estimated and the graph is interrogated to see how each cluster reshapes the clinical picture.



BDeu estimation

A weak Bayesian prior (equivalent sample size = 5) smooths rare parent configurations instead of assigning degenerate 0/1 probabilities.

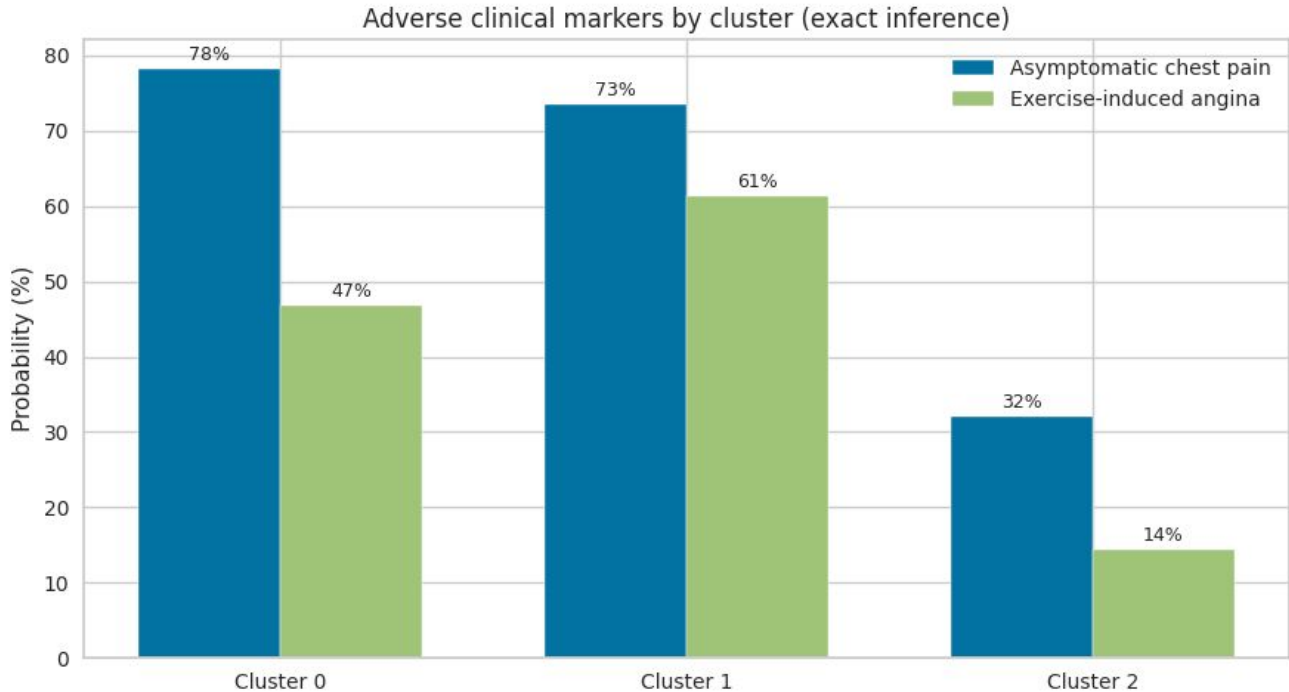


Variable Elimination

Exact inference on $P(\text{Cluster})$ and its conditionals — e.g. conditioning on the lowest vs highest cholesterol bin.

Clusters shift the clinical findings substantially away from the population marginal — behaving as distinct phenotypes, not noise.

Adverse-marker probability by cluster (exact inference)



Illustrative of the reported separation (Fig. 17) — one cluster concentrates high-burden markers, its mirror image the low-risk profile.

Which method survives millions of points?

For n samples, d dimensions, k clusters and t iterations — asymptotic cost decides real-time viability.

K-means++

Scales linearly; mini-batch streaming variant available.

$O(n k d t)$ time • $O(n d)$ memory

BEST FOR REAL-TIME

DBSCAN

$O(n \log n)$ with a spatial index in low- d ; degrades as d grows.

$O(n \log n) \rightarrow O(n^2)$ time • $O(n)$ memory

INTERMEDIATE

Agglomerative (Ward)

Needs the full pairwise matrix — offline exploration only.

$O(n^2 d + n^2 \log n)$ time • $O(n^2)$ memory

OFFLINE ONLY

Recommendation: mini-batch K-means, with PCA applied upstream to keep d small — the best balance of quality and scalability.

What the pipeline recovered



Clean foundation

Non-informative and >50%-missing columns removed; Isolation Forest filtered 5% of outliers.



Dimensionality tamed

PCA at 90% variance cut the encoded space from 19 to 10 dimensions, mitigating the curse.



Convergent structure

K-means++ & Ward agreed on $k = 3$; DBSCAN gave the cleanest partition (silhouette ~ 0.40).



Scaling is indispensable

The high unscaled silhouette was an artefact of an unnormalised cholesterol axis.



Interpretable phenotypes

A BIC-scored Bayesian Network placed Cluster as a central latent risk stratifier.



Deployable at scale

Mini-batch K-means preceded by PCA is the recommended real-time technique.

References

- [1] R. Detrano et al., "International application of a new probability algorithm for the diagnosis of coronary artery disease", American Journal of Cardiology, vol. 64, no. 5, pp. 304--310, 1989. UCI Heart Disease dataset.
- [2] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest" in Proc. 8th IEEE Int. Conf. Data Mining (ICDM), 2008, pp.413--422.
- [3] "K-Means Clustering for Heart Disease Analysis," Kaggle. [Online]. Available: <https://www.kaggle.com/competitions/k-means-clustering-for-heart-disease-analysis>
- [4] GeeksforGeeks, "Introduction to Hill Climbing --- Artificial Intelligence," GeeksforGeeks. [Online]. Available: <https://www.geeksforgeeks.org/artificial-intelligence/introduction-hill-climbing-artificial-intelligence>



Thank you

Unsupervised phenotyping recovered distinct, clinically meaningful cardiac risk profiles — from routine measurements, with no diagnostic label.

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